

THUNDERAI

AI/ML-Based Thunderstorm & Lightning Nowcasting System | SIH Problem Statement ID: SIH26072

1. Problem Statement

AI/ML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.

2. Real-World Problem

Thunderstorms and lightning can develop rapidly and may create serious risks to people, infrastructure, agriculture and public safety. Weather information is often available in complex technical forms that are difficult for ordinary users to interpret. A major challenge is providing location-specific information quickly enough for users to understand the current atmospheric situation, recognize the level of risk and take appropriate safety precautions.

3. Proposed Solution

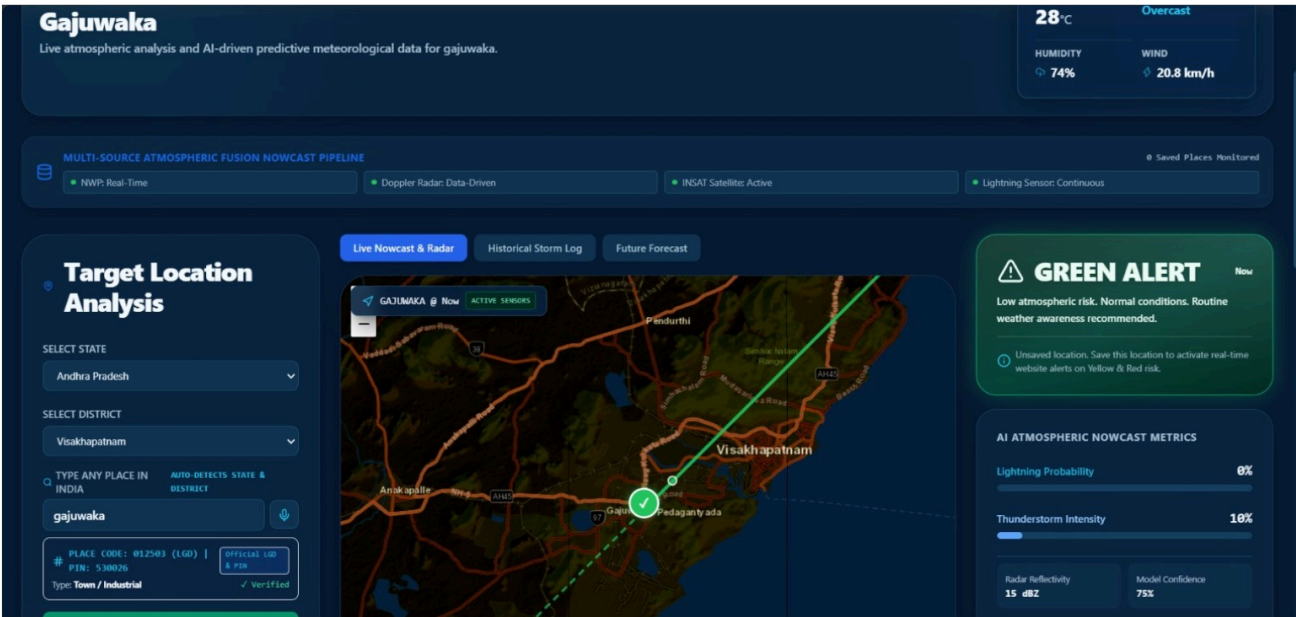
THUNDERAI is a location-focused web-based prototype designed to provide understandable and actionable information about thunderstorms and lightning conditions. Users can search for locations across India and view the selected location on an interactive map along with current meteorological information, atmospheric assessment, risk level, alerts and safety guidance. The current prototype uses available meteorological data and a rule-based atmospheric risk assessment. Future development can integrate additional real observational sources such as weather radar, satellite, lightning and numerical weather prediction data.

4. Objectives

- Provide location-specific weather and thunderstorm information.
- Allow users to search locations across India and display the selected location on an interactive map.
- Present current weather conditions in an understandable form.
- Provide thunderstorm risk assessment, alert information and clear safety precautions.
- Allow users to save important locations for monitoring and receive website notifications when alert state changes.
- Support automatic email alerts for saved locations.
- Support English, Telugu, Hindi and Tamil, together with voice-based interaction for accessibility.
- Create a foundation for future integration of radar, satellite and lightning observations.

5. Uniqueness / Innovation

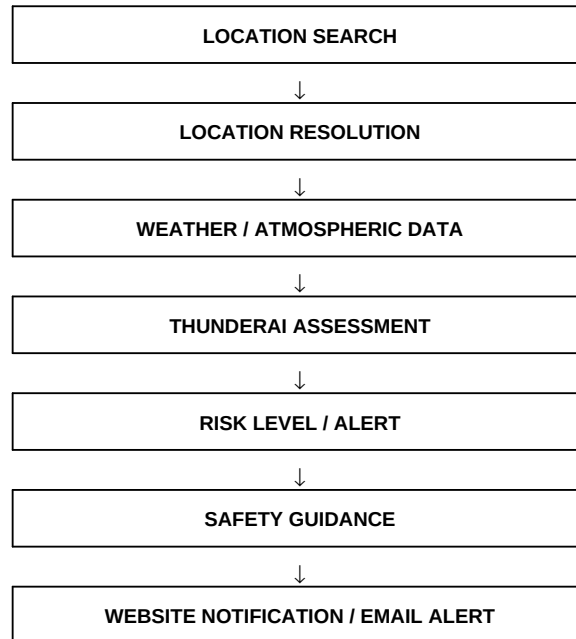
THUNDERAI focuses on making atmospheric information location-specific and understandable. Its distinctive features include India-wide location search, locality-level location resolution, interactive mapping, current weather information, atmospheric risk assessment, safety guidance and saved-location monitoring in one platform. The platform also combines website notifications and automatic email alerts for saved locations. Multilingual and voice-based interaction makes the information more accessible. The long-term innovation is to evolve the platform toward a multi-source nowcasting system that can combine radar, satellite, lightning and atmospheric model information.



System Design and Technical Approach

THUNDERAI converts a selected location and available meteorological information into an understandable assessment, alert and safety response.

6. System Architecture



The user first selects a location. The system resolves the location and obtains available meteorological information. These inputs are processed by the THUNDERAI assessment logic to determine the current risk level. The result is presented through the dashboard together with alerts and safety guidance.

7. Technical Approach

The prototype follows a location-to-assessment workflow. First, the user searches for a place. The location resolver identifies the appropriate geographic location and its associated district and state. The selected coordinates are then used to retrieve meteorological information. The current prototype uses Open-Meteo for meteorological data. Relevant weather and atmospheric variables are processed by the existing THUNDERAI assessment logic. The resulting assessment is converted into an understandable alert level and displayed on the dashboard. Safety guidance is provided according to the displayed risk condition. Saved locations are monitored by the backend so that users can receive notifications when the alert state changes.

8. Data Sources

The current prototype uses Open-Meteo for meteorological information and India geographic/location data for India-wide search and location resolution.

Current prototype: Open-Meteo provides available meteorological information such as temperature, humidity, precipitation, cloud-related information and atmospheric model variables. India geographic/location data supports location search and resolution.

Future data integration:

- Doppler weather radar observations.
- INSAT satellite observations.
- Lightning observations.
- Numerical weather prediction/model data.
- Surface meteorological observations.

9. Technology Stack

- **Frontend:** React and modern web technologies.
- **Backend:** Python with FastAPI.
- **Database:** SQLite for persistent application data such as saved-location and alert state information.
- **Weather Data:** Open-Meteo API.
- **Mapping:** Interactive web mapping using Leaflet/OpenStreetMap-based mapping.
- **Email:** Brevo email service.
- **Location Data:** India-wide geographic/location datasets and location-resolution services.



10. User Workflow

Search for a location → Location is resolved → Map moves to the selected location → Current weather is displayed → Atmospheric/risk assessment is displayed → Alert and safety guidance are shown → User can save the location → Saved location can be monitored → Website/email notification is provided when the alert state changes.

Core Features and Alert Experience

The dashboard is organized around location, current meteorological information, risk interpretation, safety guidance and saved-location monitoring.

11. Core Features

- **India-Wide Location Search:** Search across India, including cities, districts and smaller localities.
- **Interactive Map:** Automatically focuses on the selected location and displays its geographic position.
- **Current Weather:** Presents available current meteorological information such as temperature, humidity, wind and weather condition.
- **Thunderstorm Risk Assessment:** Evaluates available atmospheric information and presents a current thunderstorm risk assessment.
- **Alert Levels:** Communicates the current risk through GREEN, YELLOW and RED alert levels.
- **Safety Guidance:** Provides understandable precautions associated with hazardous weather conditions.
- **Saved Locations:** Lets users save locations they want to monitor.
- **Website Notifications:** Displays notifications for active saved-location alerts and recovery conditions.
- **Automatic Email Alerts:** The backend can monitor saved locations and send alerts to the registered user's email when the alert state changes.
- **Multilingual Support:** Supports English, Telugu, Hindi and Tamil.
- **Voice Interaction:** Supports voice interaction for location search and spoken information features.
- **Historical Log:** Provides historical meteorological information where the required data is available.
- **Future Forecast:** Provides future meteorological forecast information separately from the current thunderstorm assessment.
- **Analysis Report:** Can generate a location-specific analysis report containing relevant location, map, weather, alert and safety information.

12. Alert and Notification Workflow

Alert-state change	Notification behavior
GREEN → YELLOW	A Yellow alert notification is generated.
YELLOW → RED	A new Red alert notification is generated.
YELLOW → YELLOW	No duplicate notification is generated.
RED → RED	No duplicate notification is generated.
YELLOW / RED → GREEN	A recovery notification is generated to inform the user that the location has returned to Green conditions.

This state-based approach prevents repeated notifications for an unchanged alert condition.

13. User Experience / Workflow Explanation

The experience is designed to move from a user's simple location request to an understandable weather-risk response. After a location is searched and resolved, the map provides geographic context while the dashboard presents current weather information. The atmospheric/risk assessment is then communicated through an alert level, followed by safety guidance. A user can save a location for continued monitoring. When the saved location changes alert state, the platform can provide a website notification and, where configured, an email alert.

Example flow 1: A user searches for a locality, reviews its current weather and displayed risk level, then saves the location for monitoring.

Example flow 2: A saved location changes from Yellow to Red; the monitoring workflow generates a new Red alert notification rather than repeatedly notifying the user while the state remains unchanged.

14. Assessment Scope

The current prototype should be understood as a location-focused meteorological and rule-based assessment platform. Its long-term direction is multi-source thunderstorm and lightning nowcasting, but the intended radar, satellite and independent real-time lightning observation feeds are future integrations unless explicitly available in the current prototype.



Feasibility, Impact and Future Direction

THUNDERAI is presented as a prototype foundation that can be extended as additional observational sources and validated modelling capabilities become available.

15. Feasibility and Viability

THUNDERAI is technically feasible as a web-based platform because it uses commonly available web technologies, API-based meteorological data and a lightweight backend. The modular design allows additional data sources to be integrated in future development. The location system can support a wide range of Indian locations, while the backend can manage saved-location monitoring and notifications. The system can be expanded from a prototype into a more comprehensive atmospheric monitoring and nowcasting platform as additional real-time observational datasets become available.

16. Impact and Benefits

- **For General Users:** Provides understandable, location-specific weather risk information and safety guidance.
- **For Communities:** Can improve awareness of rapidly developing hazardous weather conditions.
- **For Disaster-Management Authorities:** The long-term system can provide additional localized information to support situational awareness.
- **For Researchers:** Provides a foundation for integrating multiple atmospheric datasets and developing validated nowcasting models.
- **For Public Safety:** Combines weather information, risk assessment, alerts and safety guidance in a single accessible platform.

17. Future Scope

- Integration of Doppler weather radar data.
- Integration of INSAT satellite observations.
- Integration of real lightning observation data.
- Integration of surface observation networks.
- Integration of additional numerical weather prediction data.
- Development and validation of ML models using real historical observations and outcomes.
- Short-term thunderstorm movement and intensity estimation.
- Improved lightning nowcasting and more localized alerts.
- Expansion of the monitoring and notification infrastructure.

The ultimate vision is to develop a validated multi-source AI/ML nowcasting system capable of providing timely and localized thunderstorm and lightning information.

18. Current Limitations

The current prototype does not have all intended operational observational feeds integrated. In particular, radar, satellite and independent real-time lightning observation feeds are not represented as fully operational inputs in the current prototype. The current assessment should therefore be understood as a prototype meteorological/rule-based assessment rather than a fully validated operational thunderstorm and lightning nowcasting model. Future ML development should use real historical observations and verified outcomes for training and validation.

19. References

- Open-Meteo — Weather and atmospheric data API.
- India Meteorological Department (IMD) — official meteorological information.
- MOSDAC / ISRO — satellite meteorological data resources.
- Relevant official disaster-management and lightning safety guidance.

